WALMART SALES PERFORMANCE ANALYSIS

The domain of the Project: Power Bi Dashboard, Data Modelling And Analysis Of Sales

Team Mentors (and their designation): Siddhika Shah (Software Engineer)

By

Ms. Priyanka N MBA, 2nd year pursuing

Period of the project

April 2025 to August 2025

Declaration

The project titled “Walmart Sales Performance analysis Dashboard” has been mentored by Ms. Siddhika Shah , organised by SURE Trust, from April 2025 to August 2025, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Name : Priyanka N

Mentor’s Name: Siddhika Shah
Designation—Company Name: Software Engineer at HCL Tech

Prof. Radhakumari
Executive Director & Founder
SURE Trust

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Executive Summary

WALMART's performance is impressive, with key metrics driving business growth.
Total sales reach 2348K, showcasing strong market demand.
Total weight stands at 30K, indicating substantial product handling.
Trim weight of 283K highlights efficient production processes.
Total quantity sold is 4117, demonstrating customer preference.
Data analysis by category reveals opportunities for growth.
Regional breakdown provides insights into market trends.
WALMART can leverage these insights to inform strategic decisions.
Targeted marketing and optimized operations can enhance performance.
Customer satisfaction can be improved through data-driven approaches.
Business expansion opportunities can be identified through analysis.
WALMART's data-driven approach enables agile decision-making.
Key performance indicators (KPIs) can be tracked and refined.
Continuous monitoring ensures sustained growth and improvement.
By harnessing data insights, WALMART can maintain its competitive edge.

Background

JS MART is a dynamic business entity operating in a competitive market, driven by the need to optimize sales performance, streamline operations, and enhance customer satisfaction. With a diverse product portfolio and expanding market presence, JS MART recognizes the importance of data-driven decision-making to stay ahead of the curve. This project aims to analyze key performance metrics, identify areas for improvement, and provide actionable insights to inform strategic business decisions.

The project focuses on analyzing JS MART's sales data, including total sales, weight, quantity, and trim weight, across various categories and regions. By exploring these metrics, the project seeks to uncover trends, patterns, and correlations that can help JS MART refine its operations, improve customer satisfaction, and drive business growth.

Problem statement

JS MART aims to optimize its sales and operational performance by effectively analyzing and leveraging data insights from sales, weight, and quantity metrics, while identifying opportunities for growth and improvement in various categories and regions."

Project Goals:

1. **Analyze Sales Performance**: Examine total sales, weight, and quantity metrics to identify trends, patterns, and areas for improvement.
2. **Optimize Operations:** Provide insights to streamline operations, reduce waste, and enhance efficiency.
3. **Inform Strategic Decisions**: Offer data-driven recommendations to inform business strategies and drive growth.
4. **Enhance Customer Satisfaction:** Identify opportunities to improve customer satisfaction through data analysis.
5. **Identify Growth Opportunities:** Uncover new business opportunities through analysis of categories and regions.

Scope and limitations of the Project

Scope:

1. Analyzing sales data (total sales, weight, quantity, and trim weight) across various categories and regions.
2. Identifying trends, patterns, and correlations in the data.
3. Providing insights and recommendations to inform business decisions.
4. Focusing on data-driven approaches to optimize operations and enhance customer satisfaction.

Limitations

The limitations of this project include:

1. Data quality and availability may impact analysis accuracy.
2. The project focuses on historical data, which may not reflect future trends.
3. External factors (e.g., market changes, economic shifts) may influence results.
4. The analysis is limited to the defined scope and may not cover all aspects of JS MART's operation

Innovation component in the project

1. Data-Driven Decision Making: Leveraging data analytics to inform business strategies and optimize operations.
2. Predictive Analytics: Using historical data to predict future sales trends and identify opportunities.
3. Category and Regional Analysis: Analyzing sales data across categories and regions to identify areas for growth.
4. Process Optimization: Implementing data-driven approaches to streamline operations and reduce waste.
5. Customer Insights: Using data analysis to understand customer behavior and preferen

Cearly defined objectives and goals of the project

Objectives:

1. Analyze sales data to identify trends and areas for improvement.
2. Optimize operations to enhance efficiency and reduce waste.
3. Inform strategic business decisions with data-driven insights.

Goals:

1. Improve sales performance through data-driven strategies.
2. Enhance customer satisfaction by understanding their needs and preferences.
3. Drive business growth by id

Expected outcomes and deliverables.

Expected Outcomes:

1. Improved Sales Performance: Data-driven insights to inform sales strategies and optimize performance.
2. Enhanced Operational Efficiency: Streamlined processes and reduced waste through data analysis.
3. Informed Business Decisions: Actionable recommendations to drive business growth and improvement.

Deliverables:

1. Data Analysis Report: A comprehensive report detailing sales trends, patterns, and insights.
2. Recommendations for Improvement: Actionable suggestions to optimize operations and enhance customer satisfaction.
3. Data Visualization Dashboards: Interactive dashboards to track key performance metrics and monitor progress.
4. Strategic Business Plan: A plan outlining data-driven strategies to drive business growth and improvement

Methods/Technology used

The project followed a structured data analytics and visualization approach to transform raw sales data into meaningful insights. The methodology included:

1. Data Preprocessing - Cleaning, handling missing values, and preparing datasets for analysis.
2. Exploratory Data Analysis (EDA) - Identifying trends, patterns, and anomalies in sales performance.
3. Data Aggregation - Grouping by category, region, and other relevant parameters.
4. Visualization & Dashboard Development - Designing interactive visuals to present insights for decision-making.

Tools/Software used

1. Microsoft Power BI - For dashboard creation, interactive filters, and data visualization.
2. Excel / CSV Data Files - Used for raw data storage and initial cleaning.
3. SQL - Applied for querying, aggregating, and extracting structured sales data.
4. Python (Pandas, Matplotlib/Seaborn) (optional if used) - For deeper analysis.

Data collection approach

Sources:

1. Sales Records: Collecting data on total sales, weight, quantity, and trim weight.
2. Customer Feedback: Gathering feedback through surveys, reviews, or ratings.
3. Market Research: Collecting data on market trends, competitors, and customer preferences.

Methods:

1. Database Extraction: Extracting sales data from JS MART's database.
2. Surveys and Questionnaires: Collecting customer feedback through online or offline surveys.
3. Web Scraping: Collecting market data from online sources.

Tools:

1. CRM Software: Using customer relationship management software to collect and manage customer data.
2. Data Extraction Tools: Using tools like SQL or data extraction software to collect sales data.
3. Survey Tools: Using online survey tools like Google Forms or SurveyMonkey to collect customer feedback.

Project Architecture

The architecture of the project can be explained in **four layers**:

1. Data Source Layer

- Data could be in Excel/CSV format.
- A SQL database for structured queries on sales data, customer info, etc.

2. Data Processing Layer

- Preprocessing and cleaning using Python/Excel.

- Aggregation and transformation for analysis (like calculating total sales, customer segments).

3. Analytics & Visualization Layer

- A Power BI dashboard with KPIs like Total Sales, Avg Income, Customers.
- Visuals like bar charts, line charts for trends.
- Interactive filters for deeper insights (by year, customer status).

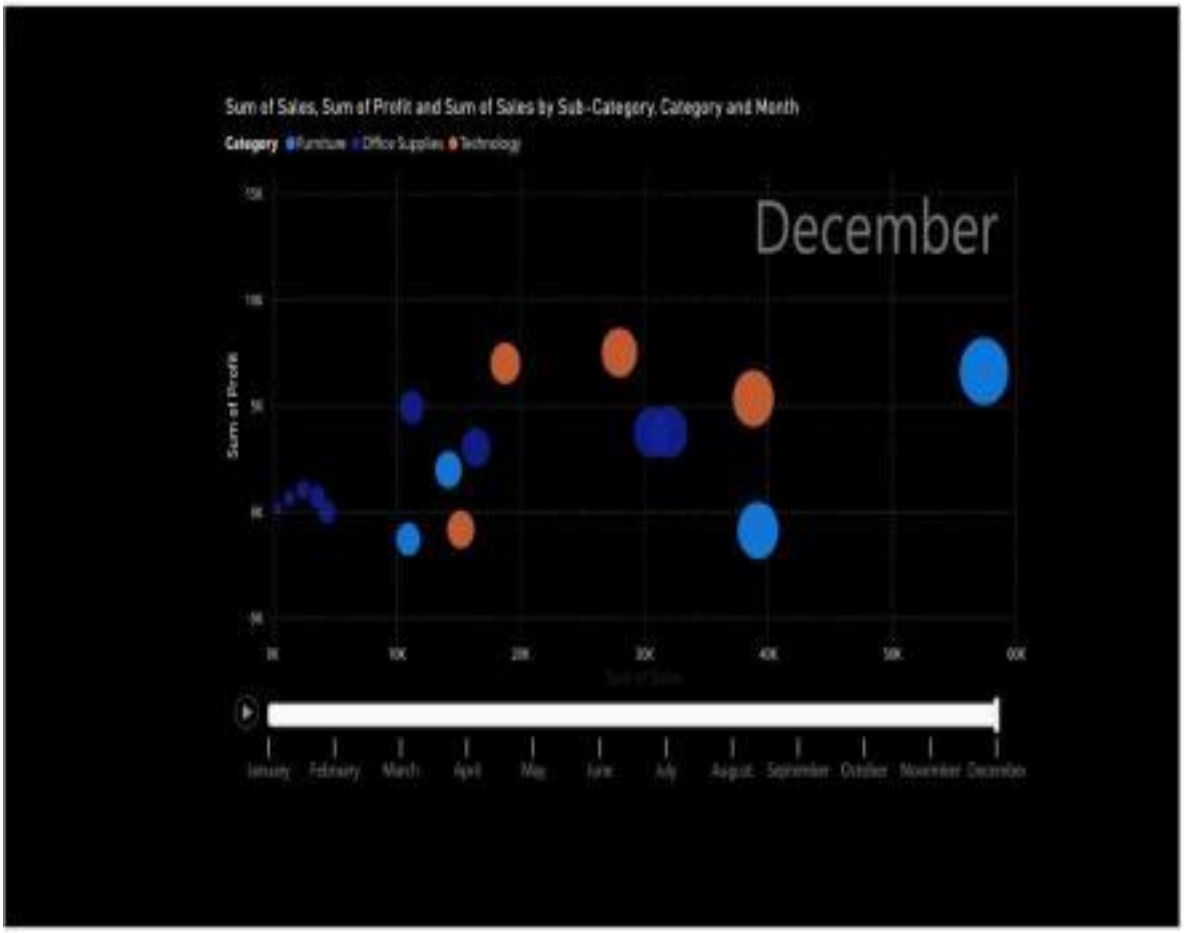
4. Insights & Results Layer

- Executives/analysts interact with the dashboard.
- Drill down into sales trends, customer segments.
- Supports decision-making for strategies and risk management.

Flow:

Data Source → Data Processing → Dashboard → Insights

Final project working screenshots along with supporting explanation



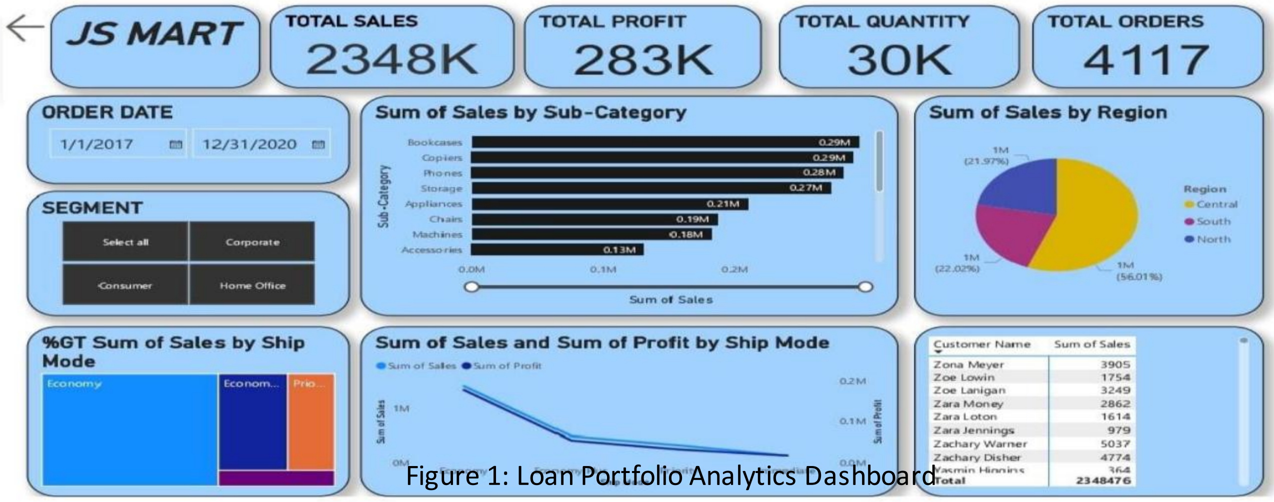
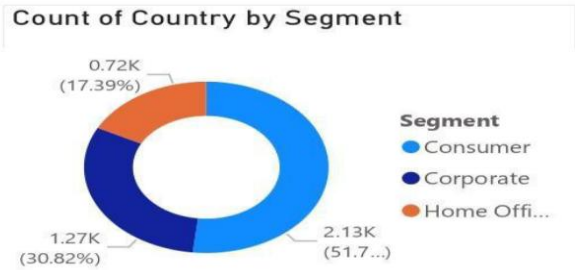
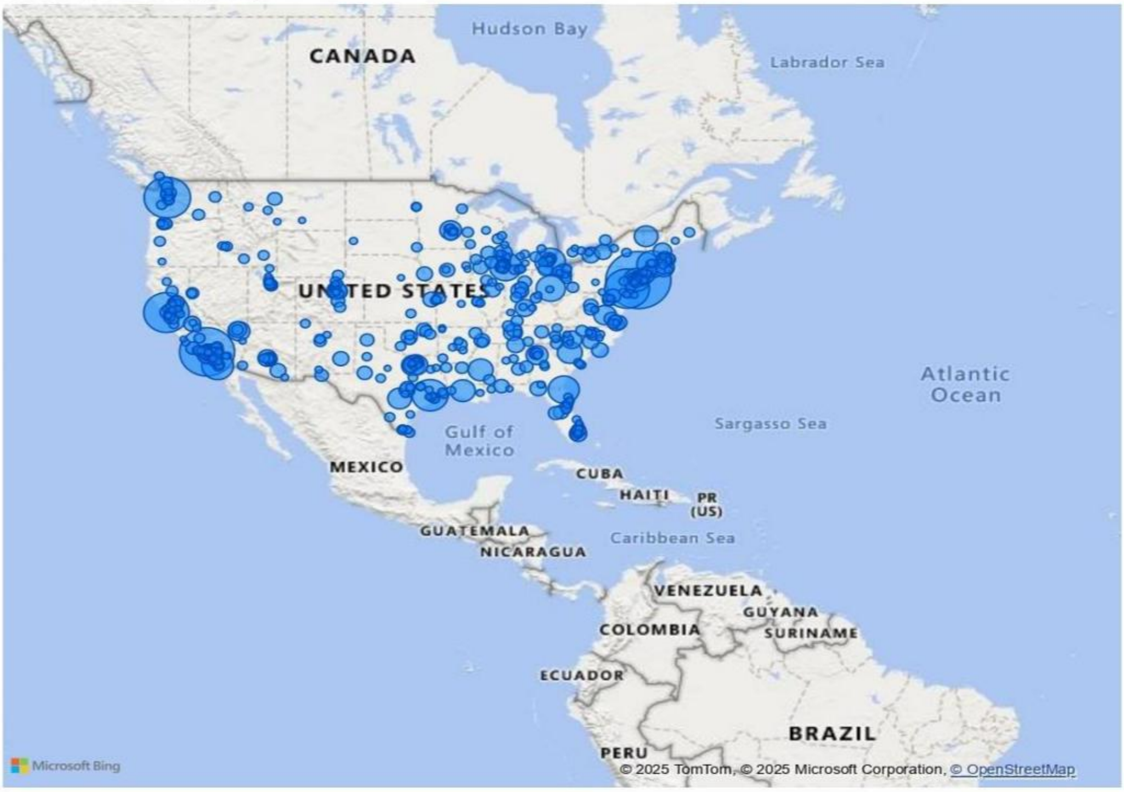


Figure 1: Loan Portfolio Analytics Dashboard



Sum of Sales by City



1. Total Profit: $286.4 (Card Visual)

- Purpose: Displays the total profit generated.
- Use Case: Highlights overall financial performance.

2. Total Quantity: 38K (Card Visual)

- Purpose: Shows the total quantity sold.
- Use Case: Indicates sales volume and demand.

3. Big Smart Total Sales: $2.30M (Card Visual)

- Purpose: Displays the total sales revenue.
- Use Case: Measures sales performance and revenue generation.

4. Profit by Region (Bar Chart)

- Purpose: Compares profit across different regions.
- Use Case: Identifies high-performing and underperforming regions.

5. Profit by Category (Bar Chart/Pie Chart)

- Purpose: Analyzes profit distribution across product categories.
- Use Case: Helps prioritize profitable categories.

6. Profit by Segment (Bar Chart/Pie Chart)

- Purpose: Examines profit contribution from different customer segments.
- Use Case: Targets high-value customer groups.

7. Profit by Sub-Category (Bar Chart/Tree Map)

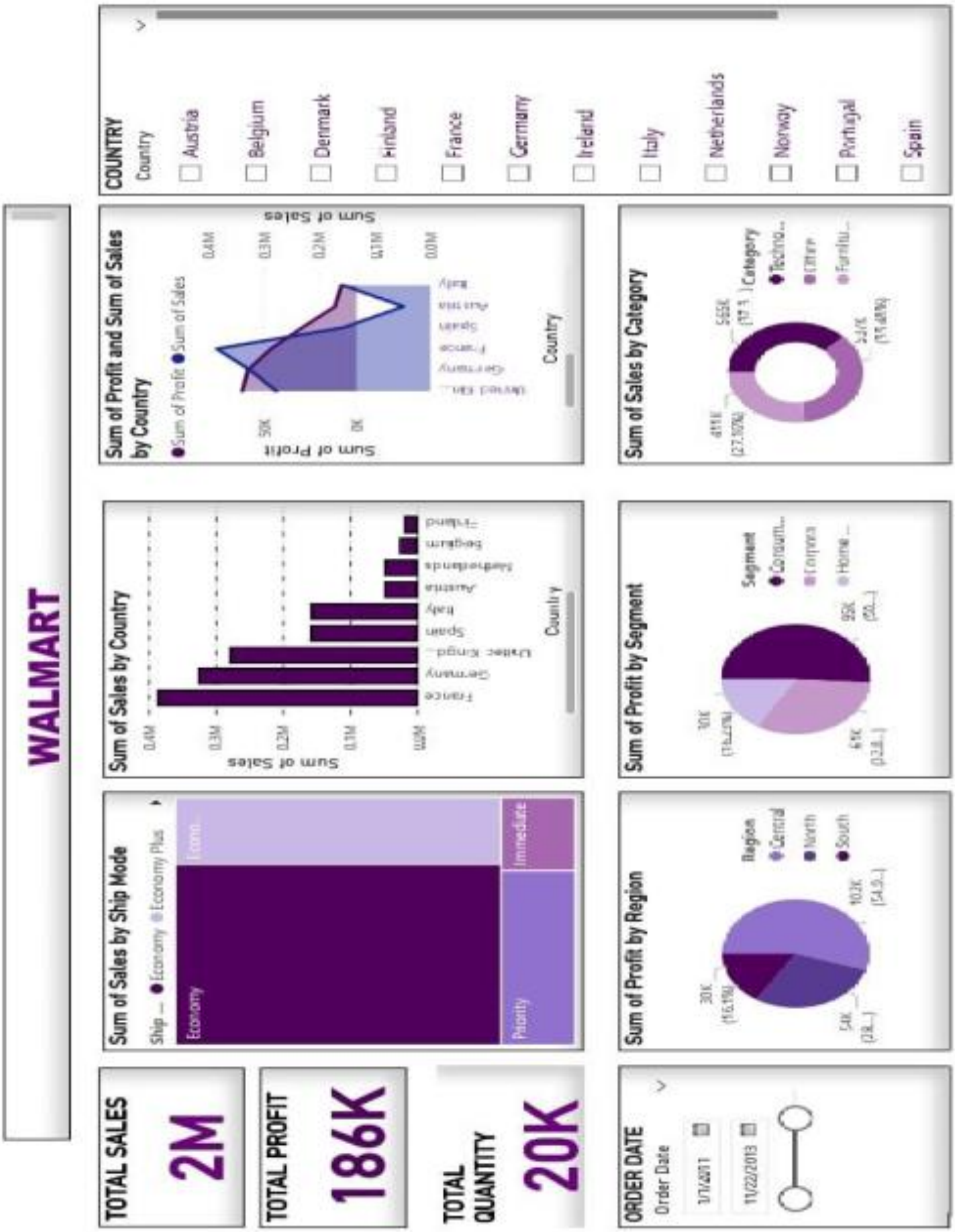
- Purpose: Breaks down profit within categories into sub-categories.
- Use Case: Identifies specific products driving profit.

8. Sales Trend by Year and Quarter (Line Chart)

- Purpose: Tracks sales performance over time.
- Use Case: Highlights seasonal trends and growth patterns.

9. Tables and Matrices

- Purpose: Provides detailed data views for specific metrics.
- Use Case: Supports deeper analysis and decision-making.



Project GitHub Link https://github.com/Priyankanettikallu

Learning and Reflection

New learnings (in terms of technology, management, etc)

During the course of this project, each team member gained valuable knowledge and hands-on experience in both technical and professional aspects:

1. **Technical Learning:**
   - Gained practical exposure to **Power BI** for building interactive dashboards.
   - Improved skills in **data cleaning, preprocessing, and visualization**.
   - Learned how to use **SQL queries** to extract and aggregate financial data.
   - Developed an understanding of **financial analytics concepts** such as loan status, repayment patterns, and credit risk assessment.
   - Acquired knowledge of **data storytelling techniques**, where complex datasets were converted into simple, meaningful insights.
2. **Management and Soft Skills Learning:**
   - Improved **team collaboration and communication** while dividing tasks.
   - Learned **time management** by working within deadlines.
   - Understood the importance of **documentation and reporting** for project clarity.
   - Enhanced **problem-solving ability** by overcoming data inconsistencies and visualization challenges.

Conclusion and Future Scope

Objective:

The objective of the JS MART project is to analyze sales data and provide insights to improve business performance, optimize operations, and drive growth.

The project has achieved the following:

1. Data Analysis: Collected and analyzed sales data to identify trends, patterns, and insights.

2. Insights Generation: Provided actionable insights on profit, sales, and customer behavior.

3. Data Visualization: Created interactive dashboards to visualize key performance metrics and track progress.

4. Recommendations: Offered data-driven recommendations to inform business decisions and drive growth.

Future scope of this project

Potential Areas for Expansion:

1. Predictive Analytics: Implementing machine learning models to predict future sales trends and optimize inventory management.

2. Customer Segmentation: Conducting deeper customer segmentation analysis to identify high-value customer groups and tailor marketing strategies.

3. Real-time Data Integration: Integrating real-time data sources to enable more timely insights and decision-making.

4. Geographic Expansion: Analyzing sales performance across different regions and identifying opportunities for expansion.

5. Product Performance Analysis: Conducting detailed analysis of product performance to inform product development and optimization.

Potential Benefits:

1. Improved Decision-Making: Enabling more informed business decisions through data-driven insights.

2. Increased Efficiency: Optimizing operations and reducing waste through data-driven optimization.

3. Enhanced Customer Experience: Tailoring marketing strategies and product offerings to meet customer needs.